

Enterprise AI-Powered Customer Analytics & Strategic Insight Framework

Churn Prediction, Revenue Forecasting, Behavioural Segmentation and LLM-Generated Executive Intelligence
for a SaaS / CRM Platform (Salesforce / Zoho style)

Attribute	Detail
Domain	SaaS / CRM Analytics / Customer Intelligence
Dataset	Star schema: 4 fact tables, 3 dimension tables
Modelling grain	5,000 customers (one row each)
Features engineered	184 modelling features
Classification target	Churn (binary)
Regression target	Next_Quarter_Revenue_USD (continuous)
Report date	29 July 2026

1. Executive Summary

This project delivers an end-to-end customer intelligence system for a subscription SaaS CRM platform. Four relational fact tables and three dimension tables were consolidated into a single customer-level table of 5,000 accounts and 184 engineered features. Three machine-learning models were trained on that table, their outputs fused into a standardised per-account intelligence profile, and a large language model used to translate those profiles into executive briefings.

The churn classifier achieves a hold-out ROC-AUC of **0.9455** and an F1 of **0.6881** on the minority churn class. The revenue model explains **96.7%** of next-quarter revenue variance with an RMSE of **\$1,363**. Segmentation resolves the portfolio into **2** behavioural clusters whose churn rates differ by a factor of roughly seven. Fusing these outputs identifies **\$711,300** of next-quarter revenue sitting in accounts with elevated churn probability.

2. Business Problem to ML Task Mapping

Every technical component is anchored to a measurable business objective, so model performance translates directly into commercial outcomes.

Business Problem	ML Task	Output	Business KPI Served
Customers cancelling subscriptions	Binary classification (XGBoost)	Churn probability 0-1	Retention rate, customer lifetime value
Unreliable revenue planning	Regression (GradientBoosting)	Next-quarter revenue (USD)	Forecast accuracy, quota setting
Undifferentiated marketing	Clustering (KMeans + PCA)	Behavioural cluster ID	Campaign conversion, personalisation
Unclear account prioritisation	Feature fusion / scoring	Risk, potential, priority scores	Revenue at risk, CSM coverage
Technical output not board-ready	LLM narrative generation	Executive briefing text	Decision latency, reporting effort

3. Methodology

3.1 Data engineering

- Fact tables aggregated from their native grain to customer grain: transactions (54,858 rows), monthly usage snapshots (131,268 rows) and engagement events (54,822 rows).
- Dimension attributes joined on Geo_ID, Industry_ID and Product_ID. Duplicate dimension keys were removed first — dim_geography ships five duplicate Geo_ID rows that would otherwise fan out the customer grain.
- 150 duplicate customer records removed, leaving 5,000 unique accounts.
- Missing values imputed with the median (numerical) and mode (categorical); zero missing cells remain.

3.2 Feature engineering

- **Tenure:** contract duration, tenure in days and years, days to contract end.
- **Usage intensity:** sessions per active user weighted by session length, DAU/MAU ratio, API calls per licence, error rate per session.
- **Engagement ratios:** events per tenure month, escalated ticket ratio, support dependency, sentiment-based engagement quality score.
- **Revenue:** quarterly revenue mean, volatility, latest value and growth ratio, average monthly revenue, CLV approximation, payment reliability.

3.3 Encoding and class balance

Binary and high-cardinality categoricals were label encoded; low-cardinality nominals were one-hot encoded, producing 190 columns. The churn target is imbalanced at roughly 11.3% positives. SMOTE was applied **only to the training split after the stratified train/test split**, so no synthetic record can reach the hold-out set.

4. Model Performance

4.1 Churn classification

Metric	Value	Business Interpretation
ROC-AUC (hold-out)	0.9455	Separates churners from retained accounts reliably
F1-score (churn class)	0.6881	Balanced precision and recall on the minority class
Baseline GradientBoosting ROC-AUC	0.9384	XGBoost selected as the stronger model
Features used	184	Post-encoding modelling features

Note: the cross-validated ROC-AUC recorded in model_metrics.json is computed over SMOTE-balanced folds and is optimistic, because synthetic minority samples leak across fold boundaries. The hold-out ROC-AUC above is the figure that should be quoted.

4.2 Explainable AI — SHAP churn drivers

Rank	Feature	Mean SHAP	Business Meaning
1	txn_count	2.8932	Transaction frequency — low volume signals disengagement
2	Renewal_Risk_Flag_Low	1.2252	CRM renewal-risk flag set to Low
3	txn_mean_payment_delay_days	0.4322	Average invoice payment delay
4	Relative_Churn_Risk_Medium	0.2398	Industry-level baseline churn risk
5	txn_renewal_count	0.2226	Number of renewal transactions completed
6	usage_mean_license_utilization_pct	0.2187	Licences actually used vs purchased
7	active_quarters	0.2054	Model-derived behavioural signal
8	Relative_Churn_Risk_Low	0.2035	Industry-level baseline churn risk
9	escalated_ticket_ratio	0.2035	Share of support tickets escalated
10	Sessions_Per_User_Day	0.1836	Model-derived behavioural signal

4.3 Revenue forecasting

Metric	Value	Business Interpretation
Selected model	gradient_boosting	Chosen over the alternative on hold-out RMSE
RMSE	\$1,363.28	Typical forecast error, penalising large misses
MAE	\$692.37	Average absolute error per account per quarter
R ²	0.9674	Share of revenue variance explained
Adjusted R ²	0.9600	Corrected for the number of features used
Cross-validated R ²	0.9557 ± 0.0105	Stability across five folds

4.4 Behavioural segmentation

Metric	Value	Assessment
Clusters selected (k)	2	Chosen by best silhouette across k = 2 to 10
Silhouette score	0.3760	Below the 0.60 project target
Davies-Bouldin index	1.1886	Lower is better; indicates moderate compactness
Calinski-Harabasz	2,723.4	Higher is better; confirms real separation

Metric	Value	Assessment
PCA components	6	90% of behavioural variance retained

Honest assessment: the 0.60 silhouette target was not met. The best achievable score was 0.3760 at k=2. SaaS behavioural features are continuous and overlapping, so accounts form a gradient rather than well-separated spheres, and no k between 2 and 10 approached the target. The clusters remain commercially useful: churn rates differ roughly sevenfold between them, which is the property the business actually needs.

5. Customer Segment Profiles

Cluster	Accounts	Mean MAU	Adoption %	Avg Monthly Rev	Churn Rate
0	1,401	46	30.2%	\$576	30.0%
1	3,599	202	73.3%	\$2,127	4.0%

Cluster 0 (*Disengaged At-Risk Accounts*) combines low usage, low adoption and heavy support dependency with a materially higher churn rate. Cluster 1 (*Loyal High-Value Accounts*) shows roughly four times the monthly active users, more than double the feature adoption, and near-negligible churn. Retention spend belongs in cluster 0; expansion and upsell motions belong in cluster 1.

6. Unified Customer Intelligence Layer

The fusion layer scores every account with all three models and emits a standardised JSON profile containing churn probability and risk band, next-quarter revenue forecast, revenue at risk, cluster assignment and label, percentile-normalised risk / potential / priority scores, and the top five per-account SHAP drivers with direction of effect.

Portfolio Indicator	Value
Accounts profiled	5,000
High-risk accounts	549
Medium-risk accounts	27
Low-risk accounts	4,424
Total forecast revenue at risk	\$711,300
Mean churn probability	11.5%
Top 5 accounts, revenue at risk	\$64,163

7. LLM-Based Executive Insight Generation

Structured intelligence profiles are passed to a Gemini model under a system instruction that forbids invented metrics and requires every claim to trace back to a supplied value. Each briefing covers account risk level, revenue impact, behavioural segment interpretation, prioritised strategic recommendations with owner and timeframe, and a targeted renewal plus upsell play. Five briefings were generated for the accounts carrying the highest forecast revenue at risk and exported to reports/executive_insights.md.

Implementation note: the originally specified gemini-pro model has been retired from the API. The pipeline probes current models at runtime and uses the first one the credential can serve.

8. Limitations and Future Work

- **Potential target leakage.** Transaction count is the dominant churn driver, but low transaction volume is partly a consequence of churning rather than a leading indicator. A production rebuild should ablate it and re-measure.
- **Optimistic cross-validation.** Cross-validation was run on SMOTE-balanced data, so synthetic samples leak across folds. Resampling should be moved inside a pipeline evaluated per fold.
- **Segmentation separation.** Silhouette of 0.376 falls short of the 0.60 target; density-based or model-based clustering may suit gradient-shaped behavioural data better.
- **Static snapshot.** The system scores a point-in-time extract. Production deployment needs scheduled re-scoring, drift monitoring and retraining triggers.

- **Insight validation.** LLM briefings are constrained by prompt design but not yet automatically checked; a factuality validator comparing generated numbers against the source profile is the natural next step.

9. Deliverables

Artefact	Location
Preprocessing pipeline	src/preprocessing.py
Model training	src/train_models.py
Intelligence fusion layer	src/fusion_layer.py
LLM insight generation	src/llm_insights.py
PDF report generator	src/generate_report.py
Executed analysis notebook	notebooks/EDA_and_Modeling.ipynb
Serialised models	models/*.pkl
Processed dataset	data/processed/processed_customer_data.csv
Customer intelligence profiles	data/processed/customer_profiles.json
Executive briefings	reports/executive_insights.md
Model metrics	reports/model_metrics.json
This report	reports/Capstone_Final_Report.pdf